

Statistical Methods I: Linear Model

Lec 02: Inference In Simple Linear Regression

Zhaoxia Yu

Professor, Department of Statistics

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Section 1

Overview

Review

- History of regression
- Simple linear regression model
- Least squares estimate
- Estimate coefficients and variance
- Coefficient of determination
- Interpretation of coefficients

Outline

- Introduction to Inference in SLR
- Hypothesis testing for regression coefficients
- Confidence intervals for regression coefficients
- Prediction (**Estimation**) and confidence intervals for mean response
- Prediction intervals for new observations
- ANOVA for SLR

Section 2

Intro

What is Statistical Inference

- Broadly speaking, statistical inference is the process of drawing conclusions about a population based on a sample of data from that population.
- There are different types of statistical inference, including:
 - **Estimation**: Estimating population parameters (e.g., mean, variance) based on sample statistics.
 - **Hypothesis Testing**: Testing assumptions or claims about population parameters using sample data.
 - **Prediction**: Making predictions about future observations based on a fitted model.
- So far, we have learned how to estimate β_0, β_1 and σ^2 in SLR. Next, we will learn how to do inference in SLR.

Inference Problems in SLR

- Hypothesis testing for regression coefficients. Example, $H_0 : \beta_1 = 0$ vs $H_a : \beta_1 \neq 0$.
- Confidence intervals for regression coefficients. Example, a 95% CI for β_1 .
- Prediction (**Estimation**) and confidence intervals for mean response. Example, estimation and a 95% CI for $E(y|x = x^*)$.
- Prediction intervals for new observations. Example, a 95% PI for a new observation y_{new} at $x = x^*$.
- ANOVA for SLR. Example, test if the regression model is useful.

How did we do inference for one-sample problems?

- To understand inference in SLR, let's first review how we did inference for one-sample problems.
- A random sample $y_1, y_2, \dots, y_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$.
- The parameter of interest: μ (unknown).
- Point estimate of μ . $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ to estimate the population mean μ .
- Sampling distribution of $\bar{y}$: $\bar{y} \sim N(\mu, \frac{\sigma^2}{n})$.
- But σ^2 is unknown. We estimate it by $s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$.
- Standard error (SE) of $\bar{y}$: $SE(\bar{y}) = \frac{s}{\sqrt{n}}$.

Inference: hypothesis testing

- Hypothesis test: $H_0 : \mu = \mu_0$ vs $H_a : \mu \neq \mu_0$.
- Test statistic:

$$t = \frac{\bar{y} - \mu_0}{SE(\bar{y})} = \frac{\bar{y} - \mu_0}{s/\sqrt{n}} \stackrel{H_0}{\sim} t_{n-1}$$

- Not normal? $y_1, y_2, \dots, y_n \stackrel{iid}{\sim} (\mu, \sigma^2)$. The results can still be used when n is large enough. Why?

Inference Confidence Interval

- A $100(1-\alpha)\%$ confidence interval for μ is given by:

$$\bar{y} \pm t_{\alpha/2, n-1} \times SE(\bar{y}) = \bar{y} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

- The standard error (SE) quantifies the precision of the estimate. The smaller the SE, the more precise the estimate.
- The critical value $t_{\alpha/2, n-1}$ determines the width of the confidence interval. A larger critical value (smaller α) results in a wider interval.

Section 3

Inference in SLR

Outline

Model: $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, $i = 1, \dots, n$ where $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$.

We will derive the following results

- 1 $E[\hat{\beta}_0] = \beta_0$, $E[\hat{\beta}_1] = \beta_1$. The LSE is **unbiased**.
- 2 $Var(\hat{\beta}_0) = \sigma^2[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}]$, $Var(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}$.
- 3 Sampling distributions of $\hat{\beta}_0, \hat{\beta}_1$.

Review: Properties of Expectation and Variance

A random variable $X \sim (\mu, \sigma^2)$, i.e.,
 $E(X) = \mu$, $Var(X) = E[(X - \mu)^2] = \sigma^2$

- $E(aX + b) = aE(X) + b = a\mu + b$
- $Var(aX + b) = a^2 Var(X) = a^2 \sigma^2$

For random variables X_1, X_2 ,

- $E(X_1 + X_2) = E(X_1) + E(X_2)$
- If X_1, X_2 are independent, then
 $Var(X_1 + X_2) = Var(X_1) + Var(X_2)$.

Expectation of $\hat{\beta}_1$

- Recall that

$$\begin{aligned}\hat{\beta}_1 &= \frac{S_{xy}}{S_{xx}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}} \\ &= \frac{\sum_{i=1}^n (x_i - \bar{x})y_i}{S_{xx}} - \frac{\bar{y} \sum_{i=1}^n (x_i - \bar{x})}{S_{xx}}\end{aligned}$$

- Since $\sum_{i=1}^n (x_i - \bar{x}) = 0$, we have

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Expectation of $\hat{\beta}_1$

- Since $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ and $E[\epsilon_i] = 0$, we have

$$E(\hat{\beta}_1) = E\left[\frac{\sum_{i=1}^n (x_i - \bar{x})y_i}{S_{xx}}\right]$$

Expectation of $\hat{\beta}_1$

- Since $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ and $E[\epsilon_i] = 0$, we have

$$\begin{aligned} E(\hat{\beta}_1) &= E\left[\frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{S_{xx}}\right] \\ &= \frac{1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) E(y_i) \end{aligned}$$

Expectation of $\hat{\beta}_1$

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Expectation of $\hat{\beta}_1$

- Since $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ and $E[\epsilon_i] = 0$, we have

$$\begin{aligned} E(\hat{\beta}_1) &= E\left[\frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{S_{xx}}\right] \\ &= \frac{1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) E(y_i) \\ &= \frac{1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) (\beta_0 + \beta_1 x_i) \\ &= \frac{\beta_1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) x_i \end{aligned}$$

Expectation of $\hat{\beta}_1$

- Since $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ and $E[\epsilon_i] = 0$, we have

$$\begin{aligned} E(\hat{\beta}_1) &= E\left[\frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{S_{xx}}\right] \\ &= \frac{1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) E(y_i) \\ &= \frac{1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) (\beta_0 + \beta_1 x_i) \\ &= \frac{\beta_1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) x_i \\ &= \beta_1 \end{aligned}$$

Variance of $\hat{\beta}_1$

- Recall that $\text{Var}(y_i) = \text{Var}(\epsilon_i) = \sigma^2$ and y_i 's are uncorrelated (a stronger assumption is independence)

$$\text{Var}(\hat{\beta}_1) = \text{Var} \left[\frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{S_{xx}} \right]$$

Variance of $\hat{\beta}_1$

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$$\begin{aligned}\text{Var}(\hat{\beta}_1) &= \text{Var} \left[\frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{S_{xx}} \right] \\ &= \frac{1}{S_{xx}^2} \text{Var} \left[\sum_{i=1}^n (x_i - \bar{x}) y_i \right]\end{aligned}$$

Variance of $\hat{\beta}_1$

- Recall that $\text{Var}(y_i) = \text{Var}(\epsilon_i) = \sigma^2$ and y_i 's are uncorrelated (a stronger assumption is independence)

$$\begin{aligned}\text{Var}(\hat{\beta}_1) &= \text{Var}\left[\frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{S_{xx}}\right] \\&= \frac{1}{S_{xx}^2} \text{Var}\left[\sum_{i=1}^n (x_i - \bar{x}) y_i\right] \\&= \frac{1}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2 \text{Var}(y_i)\end{aligned}$$

Variance of $\hat{\beta}_1$

- Recall that $\text{Var}(y_i) = \text{Var}(\epsilon_i) = \sigma^2$ and y_i 's are uncorrelated (a stronger assumption is independence)

$$\begin{aligned}\text{Var}(\hat{\beta}_1) &= \text{Var}\left[\frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{S_{xx}}\right] \\&= \frac{1}{S_{xx}^2} \text{Var}\left[\sum_{i=1}^n (x_i - \bar{x}) y_i\right] \\&= \frac{1}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2 \text{Var}(y_i) \\&= \frac{\sigma^2}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2\end{aligned}$$

Variance of $\hat{\beta}_1$

- Recall that $\text{Var}(y_i) = \text{Var}(\epsilon_i) = \sigma^2$ and y_i 's are uncorrelated (a stronger assumption is independence)

$$\begin{aligned}\text{Var}(\hat{\beta}_1) &= \text{Var}\left[\frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{S_{xx}}\right] \\&= \frac{1}{S_{xx}^2} \text{Var}\left[\sum_{i=1}^n (x_i - \bar{x}) y_i\right] \\&= \frac{1}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2 \text{Var}(y_i) \\&= \frac{\sigma^2}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2 \\&= \frac{\sigma^2}{S_{xx}},\end{aligned}$$

Section 4

Distribution

Distribution of $\hat{\beta}_1$ when σ^2 is known

- Since $\hat{\beta}_1$ is a linear combination of y_i 's and y_i 's are normally distributed, $\hat{\beta}_1$ is also normally distributed.
- Thus, when σ^2 is known, we have

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

- The standardized version:

$$Z = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\text{Var}(\hat{\beta}_1)}} = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2/S_{xx}}} = \frac{\hat{\beta}_1 - \beta_1}{\sigma/\sqrt{S_{xx}}} \sim N(0, 1)$$

CI for β_1 when σ^2 is known

- Because $\frac{\hat{\beta}_1 - \beta_1}{\sigma / \sqrt{S_{xx}}} \sim N(0, 1)$, we have

$$P\left(-z_{1-\alpha/2} \leq \frac{\hat{\beta}_1 - \beta_1}{\sigma / \sqrt{S_{xx}}} \leq z_{1-\alpha/2}\right) = 1 - \alpha$$

Therefore, a $100(1-\alpha)\%$ confidence interval for β_1 is given by:

$$\left(\hat{\beta}_1 - z_{1-\alpha/2} \frac{\sigma}{\sqrt{S_{xx}}}, \hat{\beta}_1 + z_{1-\alpha/2} \frac{\sigma}{\sqrt{S_{xx}}}\right)$$

Note $-z_{1-\alpha/2} = z_{\alpha/2}$.

Distribution of $\hat{\beta}_1$ when σ^2 is unknown

- When σ^2 is unknown, we estimate it by $s^2 = \frac{SSE}{n-2}$.
- We define $\hat{Var}(\hat{\beta}_1) = \frac{s^2}{S_{xx}}$ and the estimated standard error
 $SE(\hat{\beta}_1) = \sqrt{\hat{Var}(\hat{\beta}_1)} = \frac{s}{\sqrt{S_{xx}}}$.
- The standardized version:

$$\frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_1}{s/\sqrt{S_{xx}}}$$

How to construct a t-distributed statistic?

- Let Z be a standard normal random variable
- Let U be a chi-square random variable with ν degrees of freedom
- Suppose Z and U are independent
- Then the random variable

$$T = \frac{Z}{\sqrt{U/\nu}}$$

has a t-distribution with ν degrees of freedom, i.e., $T \sim t_\nu$.

Details about the t-statistic in SLR

The assumptions of the SLR model: 1.

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \text{where } \epsilon_1, \dots, \epsilon_n \stackrel{iid}{\sim} N(0, \sigma^2)$$

Under the assumptions, we can show

- $\hat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{S_{xx}})$, which implies that

$$Z = \frac{\hat{\beta}_1 - \beta_1}{\sigma / \sqrt{S_{xx}}} \sim N(0, 1)$$

- $\hat{\beta}_1$ and s^2 are independent, which implies that Z and s^2 are independent.
- $\frac{(n-2)s^2}{\sigma^2} = \frac{SSE}{\sigma^2} \sim \chi_{n-2}^2$.

Details about the t-statistic in SLR

- By the way to construct a t-distributed statistic, we have

$$T = \frac{Z}{\sqrt{U/(n-2)}} \sim t_{n-2}$$

where $Z = \frac{\hat{\beta}_1 - \beta_1}{\sigma/\sqrt{S_{xx}}}$ and $U = \frac{(n-2)s^2}{\sigma^2}$.

Thus, we have

$$T = \frac{\hat{\beta}_1 - \beta_1}{s/\sqrt{S_{xx}}} = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} \sim t_{n-2}$$

- Note that we have $n - 2$ degrees of freedom because we estimated two parameters β_0 and β_1 in SLR.

Section 5

CI

CI for β_1 when σ^2 is unknown

- Because $\frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} \sim t_{n-2}$, we have

$$P\left(-t_{n-2, 1-\alpha/2} \leq \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} \leq t_{n-2, 1-\alpha/2}\right) = 1 - \alpha$$

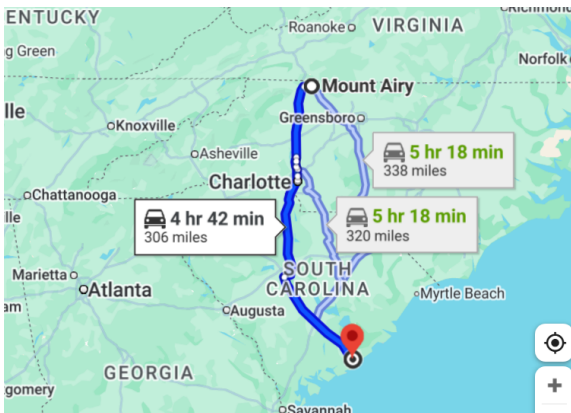
- Therefore, a $100(1-\alpha)\%$ confidence interval for β_1 is given by:

$$(\hat{\beta}_1 - t_{n-2, 1-\alpha/2} SE(\hat{\beta}_1), \hat{\beta}_1 + t_{n-2, 1-\alpha/2} SE(\hat{\beta}_1))$$

where $SE(\hat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}}$.

Example: Mount Airy and Charleston

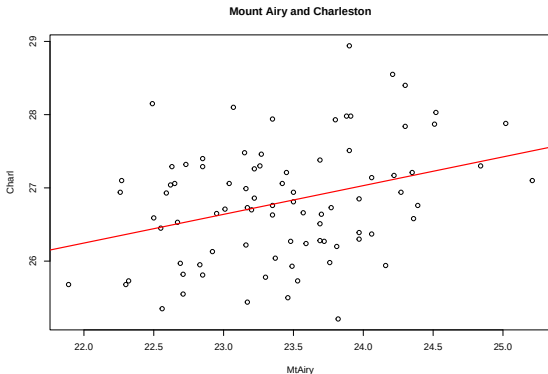
Summer temperature averages in Mount Airy, NC and Charleston, SC (1937-2021)



Example: Mount Airy and Charleston

```
mtair=read.csv("data/mta1.csv", head=TRUE)
plot(mtair$MtAiry, mtair$Charl,
     xlab="MtAiry",
     ylab="Charl",
     main="Mount Airy and Charleston")
fit <- lm(Charl ~ MtAiry, data = mtair)
abline(fit, col="red", lwd=2)
```

Example: Mount Airy and Charleston



Example: The fitting regression model

- The fitted regression model is

$$\hat{y} = 17.6 + 0.39x.$$

- Interpretation of coefficients:
 - $\hat{\beta}_0 = 17.6$: When MtAiry is 0, the expected value of Charl is 17.6. Warning: this interpretation is not meaningful because MtAiry=0 is outside the range of the data.
 - $\hat{\beta}_1 = 0.39$: For a one unit increase in MtAiry, the expected change in Charl is an increase of approximately 0.39 units.

Example: Mount Airy and Charleston

```
##  
## Call:  
## lm(formula = Charl ~ MtAiry, data = mtair)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -1.7502 -0.6291 -0.0123  0.5585  1.9484   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  17.6208     2.8325   6.221 1.91e-08 ***  
## MtAiry        0.3921     0.1209   3.244 0.0017 **    
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.7581 on 83 degrees of freedom  
## Multiple R-squared:  0.1125, Adjusted R-squared:  0.1018   
## F-statistic: 10.52 on 1 and 83 DF,  p-value: 0.001701
```

Example: Mount Airy and Charleston

To obtain a 95% CI for β_1 , we need the following values from the R output

- $\hat{\beta}_1 \approx 0.39$
- $se(\hat{\beta}) \approx 0.1209$
- $n - 2 = 83$ (degrees of freedom) and $t_{83,0.975} \approx 1.99$.
- Therefore, a 95% CI for β_1 is given by:

$$(0.39 - 1.99 \times 0.1209, 0.39 + 1.99 \times 0.1209) = (0.15, 0.63)$$

```
c(0.39 - qt(0.975, 83)* 0.1209, 0.39 + qt(0.975, 83)* 0.1209)
## [1] 0.1536919 0.6263081
```

Example: Mount Airy and Charleston

We can also use the R function `confint()` to obtain the confidence intervals for regression coefficients.

```
confint(fit, level=0.95)
```

```
##                2.5 %      97.5 %  
## (Intercept) 11.9870442 23.2545143  
## MtAiry      0.1516618 0.6325081
```

```
confint(fit, "MtAiry", level=0.95)
```

```
##                2.5 %      97.5 %  
## MtAiry 0.1516618 0.6325081
```

Therefore: $\hat{\beta}_1 \approx 0.39$ and a 95% CI for β_1 is (0.15, 0.63).

Interpretation of the 95% CI for β_1 : With 95% confidence, the estimated average summer temp in Charleston increases between 0.15 and 0.63 degrees Celsius per degree Celsius increment in average summer temp in Mount Airy.

Section 6

Hypothesis Testing

Hypothesis Testing in SLR

- We will focus on hypothesis testing for β_1 .
- In SLR, we are often interested in testing whether there is a linear relationship between x and y .
- The **null** hypothesis is that there is no linear relationship, i.e., $\beta_1 = 0$.
- The **alternative** hypothesis is that there is a linear relationship, i.e., $\beta_1 \neq 0$.
- Thus, we want to test the following hypotheses:

$$H_0 : \beta_1 = 0 \quad \text{vs} \quad H_a : \beta_1 \neq 0$$

Hypothesis testing in general: Five steps

- Step 1: State the null and alternative hypotheses.
- Step 2: Choose a significance level α .
- Step 3: Calculate a test statistic using the sample.
- Step 4: Determine the rejection region or p-value.
- Step 5: Make a decision and interpret the results.

Test Statistic for β_1

- The test statistic is given by:

$$T = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} = \frac{\hat{\beta}_1}{s/\sqrt{S_{xx}}}$$

- When H_0 is true, i.e., $\beta_1 = 0$, $T \sim t_{n-2}$.

Rejection/critical Region and p-value

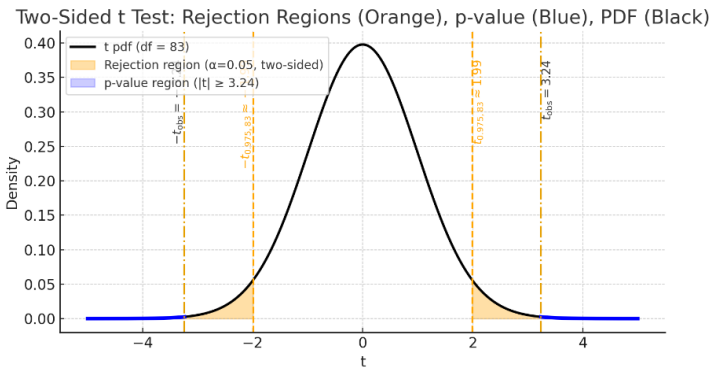
- Rejection region approach: Reject H_0 if $|T| > t_{n-2, 1-\alpha/2}$.
- p-value approach: The p-value (two-sided) is given by

$$p - value = 2P(T \geq |t_{obs}|)$$

where t_{obs} is the observed value of the test statistic T . What is the distribution of T ?

- If $p - value < \alpha$, we reject H_0 .
- If $p - value \geq \alpha$, we fail to reject H_0 . (there is not enough evidence to reject H_0).

Rejection/critical Region and p-value



Example: Mount Airy and Charleston

- Recall that $\hat{\beta}_1 = 0.39$ and $SE(\hat{\beta}_1) = 0.1209$.
- The t-statistic is $T = \frac{0.3921 - 0}{0.1209} \approx 3.24$.
- The degrees of freedom is $n - 2 = 83$.
- The critical value $t_{83,0.975} \approx 1.99$. In R, use “qt(0.975, 83)”.
We reject the null hypothesis because $|3.24| > 1.99$.
- The p-value is given by $2P(T \geq 3.24) \approx 0.0017$ (using R).
We reject the null hypothesis because $0.0017 < 0.05$.
- We conclude that there is a significant linear relationship between the average summer temperatures in Mount Airy and Charleston.

Example: Mount Airy and Charleston

```
##  
## Call:  
## lm(formula = Charl ~ MtAiry, data = mtair)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -1.7502 -0.6291 -0.0123  0.5585  1.9484   
##  
## Coefficients:  
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## F-statistic: 10.52 on 1 and 83 DF,  p-value: 0.001701
```

One-sided tests: $H_1 : \beta_1 > 0$

- We can also perform one-sided tests. For example, to test

$$H_0 : \beta_1 \leq 0 \quad \text{vs} \quad H_a : \beta_1 > 0$$

- Rejection region approach: Reject H_0 if $T > t_{n-2, 1-\alpha}$.
- p-value approach: The p-value (right-sided) is given by

$$p - \text{value} = P(T \geq t_{obs})$$

We reject H_0 if $p - \text{value} < \alpha$.

One-sided tests: $H_1 : \beta_1 > 0$

- In some applications, we may be interested in testing one-sided alternatives. For example, we would like to know whether Mount Airy has a positive linear relationship with Charleston, i.e.,

$$H_0 : \beta_1 \geq 0 \quad \text{vs} \quad H_a : \beta_1 < 0$$

- Rejection region approach: Reject H_0 if $T < -t_{n-2, 1-\alpha}$.
- p-value approach: The p-value (left-sided) is given by

$$p\text{-value} = P(T \leq t_{obs})$$

Example: Mount Airy and Charleston

- For the Mount Airy and Charleston example, we have $t_{obs} = 3.24$ and $n - 2 = 83$.
- The critical value $t_{83,0.95} \approx 1.66$. We reject the null hypothesis because $3.24 > 1.66$.
- The p-value is given by $P(T \geq 3.24) \approx 0.00085$ (using R). We reject the null hypothesis because $0.00085 < 0.05$.

Example: Mount Airy and Charleston

